

**DESIGN AND DEVELOPMENT OF A CONTEXT-AWARE SENTIMENT
ANALYSIS SYSTEM USING MULTI-LEVEL NLP**

ASSESSMENT REPORT

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GROUP 2 - MULTI-LEVEL NLP SENTIMENT ANALYSIS

DESIGN AND DEVELOPMENT OF A CONTEXT-AWARE SENTIMENT ANALYSIS SYSTEM USING MULTI-LEVEL NLP

GROUP 2 - TF-IDF + LOGISTIC REGRESSION MULTI-LEVEL NLP
APPROACH

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1. Problem Formulation

Online customer reviews frequently contain multiple opinions within the same review. A review may contain positive opinions about one product aspect and negative opinions about another aspect. Simple sentiment analysis systems that classify text by counting positive and negative words may therefore produce incorrect results.

The problem considered in this assignment is the development of a Python-based context-aware sentiment analysis system that classifies a customer review as Positive, Negative, or Neutral by considering linguistic information at multiple levels. Group 2 uses the same problem statement as Group 1, but replaces the core rule-based final decision mechanism with a supervised machine-learning classifier based on TF-IDF features and Logistic Regression.

Customer Review Used for Analysis:

"I was expecting the phone to be amazing. The display is bright and the camera takes excellent pictures, but the battery is disappointing. It does not last long, and I am unhappy with it. However, the service team replaced it quickly, which was impressive. Overall, I am satisfied with the phone."

Proposed Group 2 solution: a TF-IDF + Logistic Regression multi-level NLP system. The classifier receives lexical features from unigrams, bigrams, and trigrams together with structured linguistic features derived from morphology, POS information, negation, syntax, semantics, WSD, reference resolution, discourse markers, aspect evidence, and explicit overall sentiment.

Input

The input to the system is a natural-language customer review containing opinions about a product.

Output

1. Morphological analysis
2. N-gram representations
3. POS information
4. Syntactic/CFG analysis
5. Negation detection
6. Semantic representation
7. Word Sense Disambiguation
8. Reference resolution
9. Discourse relations
10. Product aspects
11. Aspect-level sentiment
12. TF-IDF feature representation
13. Machine-learning sentiment prediction
14. Overall sentiment score/evidence
15. Natural-language explanation

16. Tamil translation of the explanation

Problem Question

How can the sentiment of a customer review be correctly determined when positive and negative opinions, negation, contrast, references, and multiple product aspects occur in the same text, while using a learned classifier rather than a purely rule-based final decision?

Why Keyword-Based Analysis Is Insufficient

- "does not last long" expresses a negative opinion through negation.
- "but" changes the discourse relationship between camera/display and battery.
- "However" introduces a contrasting positive opinion about service.
- "it" refers to different entities depending on context.
- "Overall" indicates a global/final opinion.
- Different aspects of the same product have different sentiment polarities.

2. Objectives

17. To develop a multi-level NLP system for context-aware sentiment analysis.
18. To preprocess and tokenize customer reviews.
19. To perform morphological analysis and stemming.
20. To generate unigram, bigram, and trigram representations.
21. To perform POS-based linguistic analysis.
22. To construct CFG rules for syntactic analysis.
23. To explicitly detect and handle negation.
24. To represent selected statements using First-Order Predicate Calculus.
25. To perform basic Word Sense Disambiguation using contextual rules.
26. To resolve references such as "it" and "which".
27. To identify discourse relations such as contrast and conclusion.
28. To identify product aspects such as display, camera, battery, service, and phone.
29. To calculate aspect-level sentiment evidence.
30. To convert textual evidence to TF-IDF features.
31. To train a Logistic Regression sentiment classifier.
32. To combine lexical and structured linguistic features for prediction.
33. To classify the review as Positive, Negative, or Neutral.
34. To generate a natural-language explanation of the result.
35. To translate the generated explanation into Tamil.
36. To demonstrate why multi-level machine-learning analysis is more reliable than simple keyword counting.

3. Expected Outcomes

- Correct tokenization of the review.
- Successful stemming of selected words.
- Generation of N-grams.

- Identification of relevant POS information.
- Construction of suitable CFG rules.
- Identification of negative expressions and negation scope.
- Recognition of positive and negative aspects.
- Resolution of contextual references.
- Identification of discourse relations.
- Aspect-level sentiment analysis.
- TF-IDF feature extraction.
- Logistic Regression prediction.
- Overall sentiment classification.
- Generation of an explanation supporting the classification.
- Tamil translation of the generated explanation.

Aspect	Expected Sentiment
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Expected final classification: OVERALL SENTIMENT = POSITIVE

4. Requirements

4.1 Functional Requirements

37. Accept a customer review as input.
38. Perform text preprocessing.
39. Tokenize the input.
40. Perform morphological analysis.
41. Apply stemming.
42. Generate unigrams, bigrams, and trigrams.
43. Identify word classes and POS information.
44. Apply CFG-based syntactic analysis.
45. Detect explicit negation.
46. Determine the scope of negation.
47. Generate semantic representations for selected statements.
48. Perform Word Sense Disambiguation.
49. Resolve contextual references.
50. Identify discourse markers.
51. Identify product aspects.
52. Determine sentiment associated with each aspect.
53. Extract TF-IDF textual features.
54. Train and apply Logistic Regression.

55. Generate overall sentiment prediction.
56. Generate a natural-language explanation.
57. Provide a Tamil translation.

4.2 Software Requirements

- Python 3.x
- NLTK
- scikit-learn
- SciPy sparse matrices
- Regular Expression library (re)
- VS Code / Jupyter Notebook / Google Colab
- Optional translation library/API

4.3 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Basic processor capable of running Python
- Internet connection if an external translation service is used

5. Constraints

58. The final result should not depend only on keyword frequency.
59. Negation must be handled explicitly.
60. Contrast markers must influence interpretation.
61. Aspect-level and overall sentiment must be distinguished.
62. Ambiguous references must be handled using contextual rules.
63. The system should provide intermediate outputs.
64. The classifier is demonstrated with a small academic training set rather than a production-scale corpus.
65. The prototype is primarily designed for English customer reviews.
66. The manually prepared training examples and sentiment lexicon do not cover every possible expression.
67. The system is an academic prototype rather than a commercial deployment.

6. Assumptions

68. The input review is written in English.
69. The review is grammatically understandable.
70. Product aspects can be identified using predefined aspect terms and contextual rules.
71. A small manually prepared labeled dataset is sufficient for demonstrating the ML pipeline.
72. The sentiment-bearing expressions in the sample have identifiable polarity.
73. Discourse markers such as "but", "However", and "Overall" provide useful contextual information.
74. Pronouns can be resolved using nearby entities and discourse context.
75. The explicit overall statement is an important feature for review-level classification.

76. The classifier is intended to demonstrate the method, not to claim benchmark performance.
 77. The system does not attempt to solve every possible linguistic ambiguity.

7. Application of All Five Units

7.1 Unit I - Morphology

Morphological processing identifies the basic or stemmed form of words. Group 2 retains the same morphological stage used in the source problem, but the resulting stem information is also treated as a feature family for the learned classifier.

Word	Stem
amazing	amaz
pictures	pictur
disappointing	disappoint
unhappy	unhappi
replaced	replac
impressive	impress
satisfied	satisfi

The system uses PorterStemmer from NLTK. Regular expressions are used for preprocessing, punctuation removal, and identification of specific linguistic patterns.

Conceptual finite-state morphological processing: Input Word -> Lexical Form -> Morphological Rule -> Stem.

7.2 Unit II - N-Grams and POS

Group 2 uses unigrams, bigrams, and trigrams as the principal lexical features. Unlike a pure keyword counter, the TF-IDF vectorizer gives higher or lower importance to terms according to their distribution in the small training corpus.

- Unigrams: display, bright, camera, excellent, battery, disappointing, service, impressive, satisfied.
- Bigrams: display bright, excellent pictures, battery disappointing, does not, last long, service team.
- Trigrams: camera takes excellent, does not last, I am unhappy, replaced it quickly, Overall I am.

Word	Role
display	NOUN
bright	ADJECTIVE
camera	NOUN
excellent	ADJECTIVE
battery	NOUN
disappointing	ADJECTIVE
service	NOUN
impressive	ADJECTIVE
satisfied	ADJECTIVE
it	PRONOUN

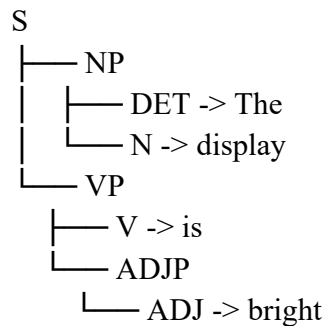
The structured POS information is not used as a replacement for the classifier. It is used to explain relationships such as noun-aspect + adjective-opinion and to derive additional feature values.

7.3 Unit III - Context-Free Grammar

CFG rules are used to represent sentence structure and to identify the relation between product aspects and predicates.

S → NP VP
 VP → V NP
 VP → V ADJP
 NP → DET N
 NP → DET N PP
 PP → PREP NP
 ADJP → ADV ADJ
 ADJP → ADJ

For "The display is bright":



CFG analysis helps create interpretable structured features, such as the presence of an adjective predicate attached to a product noun.

7.4 Negation Scope

The phrase "It does not last long" contains the explicit negation marker "not". The system detects the whole expression "does not last long" and generates a negative structured feature for battery performance. This feature is added to the ML input rather than making the final decision by rule.

7.5 Unit IV - Semantic Analysis

Selected statements are represented using simplified First-Order Predicate Calculus (FOPC).

Sentence	Representation
The display is bright.	Bright(Display)
The camera takes excellent pictures.	Excellent(CameraPictures)
The battery is disappointing.	Disappointing(Battery)
The service team replaced it quickly.	Replaced(ServiceTeam, Product)
I am satisfied with the phone.	Satisfied(User, Phone)

7.6 Word Sense Disambiguation

- bright -> interpreted as display brightness because it modifies the display.
- service -> interpreted as customer/service support because it occurs in "service team" and a replacement event.
- phone -> interpreted as the mobile device/product.

These contextual sense labels become explanatory features and can also contribute to aspect mapping.

7.7 Unit V - Reference Resolution

Expression	Resolved Meaning
first 'it'	battery/product context
second 'it'	product/item being replaced
'which'	the preceding service replacement event

Reference resolution is handled with simple context rules. The result is converted into explicit feature signals so the model can distinguish statements about battery performance from statements about service replacement.

7.8 Discourse Analysis

Marker	Relation	Effect
but	Contrast	Separates positive display/camera opinions from negative battery opinion
However	Contrast	Introduces positive service information
Overall	Global assessment	Introduces final user opinion

8. Proposed Solution

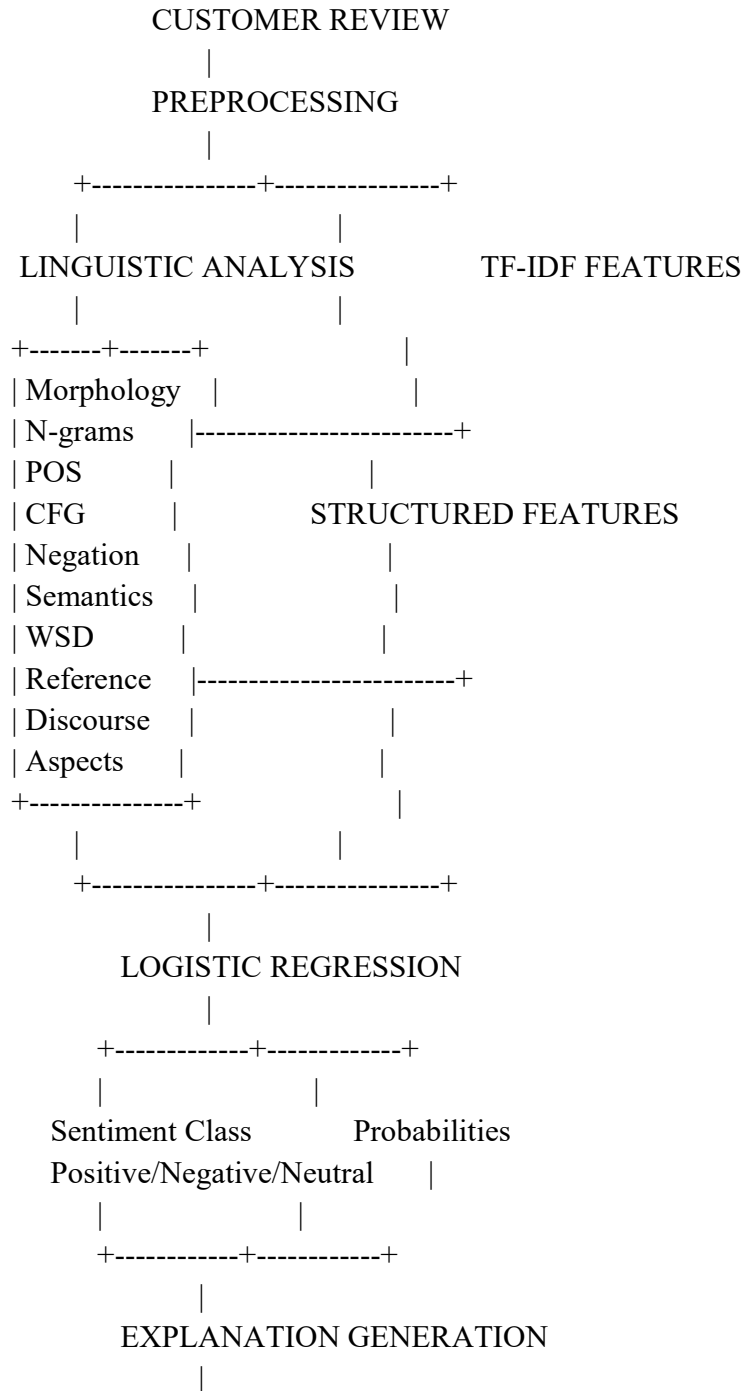
Group 2 proposes a supervised machine-learning solution based on TF-IDF feature extraction and Logistic Regression. The linguistic pipeline remains multi-level so that the same PS is addressed comprehensively.

- Regular expressions
- Sentence segmentation and tokenization
- Porter stemming
- Unigram/bigram/trigram features
- POS information
- CFG-based syntactic analysis
- Negation detection
- Semantic/FOPC representations
- WSD rules
- Reference resolution
- Discourse analysis
- Aspect extraction
- Structured sentiment evidence

- TF-IDF vectorization
- Logistic Regression
- Natural-language generation
- Tamil translation

Core algorithm distinction: Group 1 uses a hand-written rule-based final score. Group 2 uses a learned Logistic Regression classifier over TF-IDF and structured linguistic features.

9. System Architecture



TAMIL TRANSLATION

10. Methodology

Step 1 - Preprocessing

The review is divided into sentences and tokens. Important markers such as not are preserved.

Step 2 - Morphological Analysis

Porter stemming is applied to selected tokens.

Step 3 - N-Gram Generation

Unigrams, bigrams, and trigrams are created.

Step 4 - POS Analysis

Basic word classes are identified using a lightweight rule-based fallback.

Step 5 - CFG Analysis

Selected sentences are represented with context-free grammar rules.

Step 6 - Negation Detection

Patterns such as does not + verb and is not + adjective are detected.

Step 7 - Semantic Analysis

Key statements are mapped to simplified predicate representations.

Step 8 - WSD

Contextual rules assign likely meanings to ambiguous terms.

Step 9 - Reference Resolution

Pronouns and discourse references are linked to relevant entities.

Step 10 - Discourse Analysis

but, However, and Overall are extracted as structural markers.

Step 11 - Aspect Identification

display, camera, battery, service, and phone are used as predefined aspects.

Step 12 - Structured Feature Construction

Counts and flags for positive evidence, negative evidence, negation, contrast, and explicit overall satisfaction are produced.

Step 13 - TF-IDF Feature Extraction

The review is vectorized with unigram, bigram, and trigram TF-IDF weights.

Step 14 - Feature Fusion

Sparse TF-IDF features are concatenated with the structured linguistic feature vector.

Step 15 - Logistic Regression

A multinomial-capable Logistic Regression model predicts Positive, Negative, or Neutral.

Step 16 - Aspect Analysis

Independent aspect scores are generated for explanation and diagnostic reporting.

Step 17 - Explanation Generation

The system creates a natural-language explanation from aspect and discourse evidence.

Step 18 - Tamil Translation

The explanation is translated into Tamil using a predefined output for the academic demonstration.

11. Algorithm

Algorithm: TF-IDF + Logistic Regression with Multi-Level Linguistic Feature Engineering

78. Read the review.
79. Split the review into sentences.
80. Tokenize and normalize the review.
81. Apply stemming.
82. Generate unigrams, bigrams, and trigrams.
83. Identify POS information.
84. Apply CFG rules to selected sentences.
85. Detect negation and its scope.
86. Generate semantic/FOPC representations.
87. Perform contextual WSD.
88. Resolve references such as it and which.
89. Detect discourse markers.
90. Identify product aspects.
91. Generate structured sentiment/context features.
92. Load the small labeled training corpus.
93. Fit a TF-IDF vectorizer on the training texts.
94. Transform training texts and the input review to TF-IDF vectors.
95. Concatenate TF-IDF vectors with structured linguistic features.
96. Train Logistic Regression on the fused training representation.
97. Predict the sentiment class and class probabilities for the review.
98. Calculate aspect-level evidence scores for interpretability.
99. Generate a natural-language explanation.
100. Translate the explanation into Tamil.
101. Display intermediate outputs and final classification.

12. Pseudocode

BEGIN

INPUT review

```

sentences <- split_into_sentences(review)
tokens <- tokenize(review)
stems <- stem(tokens)
unigrams <- generate_unigrams(tokens)
bigrams <- generate_bigrams(tokens)
trigrams <- generate_trigrams(tokens)
pos_tags <- perform_POS_analysis(review)
cfg_result <- perform_CFG_analysis(selected_sentences)
negation <- detect_negation(review)
semantic_forms <- generate_FOPC(selected_sentences)
word_senses <- perform_WSD(review)
references <- resolve_references(review)
discourse <- detect_discourse_markers(review)
aspects <- identify_aspects(review)

structured_features <- build_structured_features(
  review, negation, pos_tags, discourse, aspects, references
)

train_texts, train_labels <- load_labeled_training_examples()
vectorizer <- fit_TFIDF(train_texts, ngram_range=(1,3))
X_text <- transform(vectorizer, train_texts)
X_struct <- build_structured_features_for_training(train_texts)
X_train <- concatenate(X_text, X_struct)

model <- train_Logistic_Regression(X_train, train_labels)

X_review_text <- transform(vectorizer, review)
X_review_struct <- structured_features
X_review <- concatenate(X_review_text, X_review_struct)

prediction <- model.predict(X_review)
probability <- model.predict_proba(X_review)

aspect_scores <- calculate_aspect_scores(review)
explanation <- generate_explanation(prediction, aspect_scores, discourse, negation)
tamil_output <- translate_to_tamil(explanation)

DISPLAY all intermediate results
DISPLAY prediction
DISPLAY probability
DISPLAY aspect_scores

```

DISPLAY explanation
 DISPLAY tamil_output

END

13. Implementation

13.1 Environment and Tools

Component	Tool
Programming Language	Python 3
Regular Expressions	Python re
Tokenization	NLTK / regex
Stemming	NLTK PorterStemmer
N-Grams	NLTK + scikit-learn TF-IDF
POS Analysis	Rule-based fallback
CFG	NLTK CFG / explicit grammar rules
Sentiment Processing	TF-IDF + Logistic Regression + structured features
Sparse Feature Fusion	SciPy hstack
Translation	Translation API/library or predefined demonstration output
Development Environment	VS Code / Jupyter / Google Colab

13.2 Training Data for Academic Demonstration

Because the original PS provides one target review, the prototype uses a small manually prepared training set containing representative positive, negative, and neutral examples. This is sufficient to demonstrate the algorithmic pipeline, but it is not a substitute for a large validated sentiment dataset.

Class	Example
Positive	The phone is amazing and excellent.
Positive	The display is bright and clear.
Positive	I am very satisfied with this phone.
Negative	The battery is disappointing and the phone is poor.
Negative	The display is dull and the camera is terrible.
Negative	I am unhappy with the battery and the phone.
Neutral	The phone has a 6.5-inch display and 128 GB storage.
Neutral	The device weighs 180 grams and has 128 GB storage.

13.3 Python Source Code

```
import re
from collections import defaultdict

from nltk.stem import PorterStemmer
from nltk.util import ngrams
```

```

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from scipy.sparse import hstack, csr_matrix

# -----
# 1. INPUT REVIEW
# -----
review = """
I was expecting the phone to be amazing. The display is bright and the
camera takes excellent pictures, but the battery is disappointing.
It does not last long, and I am unhappy with it. However, the service
team replaced it quickly, which was impressive. Overall, I am satisfied
with the phone.
"""

# -----
# 2. SMALL LABELED TRAINING SET
# -----
train_texts = [
    "The phone is amazing and excellent.",
    "The display is bright and clear.",
    "I am very satisfied with this phone.",
    "The camera takes excellent pictures and the service is impressive.",
    "Overall I am happy and satisfied.",
    "The phone works perfectly and I love the camera.",
    "The battery is disappointing and the phone is poor.",
    "The display is dull and the camera is terrible.",
    "I am unhappy with the battery and the phone.",
    "The service was disappointing and slow.",
    "Overall I hate this phone and it is bad.",
    "The battery does not last long and performance is poor.",
    "The phone has a 6.5-inch display and 128 GB storage.",
    "The device weighs 180 grams and has 128 GB storage.",
    "The phone includes a USB port and a 6.5-inch screen."
]

# 1 = Positive, -1 = Negative, 0 = Neutral
train_labels = [1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, 0, 0, 0]

# -----
# 3. SENTIMENT LEXICON FOR STRUCTURED FEATURES
# -----
positive_words = {

```



```

    "amazing": 3, "bright": 2, "excellent": 3,
    "impressive": 3, "satisfied": 3, "quickly": 1,
    "happy": 2, "perfectly": 3, "love": 3
}

negative_words = {
    "disappointing": -3, "unhappy": -3, "poor": -3,
    "dull": -2, "terrible": -3, "hate": -3,
    "bad": -3, "slow": -1
}

def structured_features(text):
    """Create interpretable numeric context features."""
    t = text.lower()

    positive_hits = sum(t.count(word) for word in positive_words)
    negative_hits = sum(t.count(word) for word in negative_words)

    negation_patterns = [
        r"\bdoes not\s+\w+(?:\s+\w+)?\b",
        r"\bdo not\s+\w+\b",
        r"\bis not\s+\w+\b",
        r"\bnot\s+\w+\b"
    ]
    negation_count = sum(len(re.findall(p, t)) for p in negation_patterns)

    contrast_count = t.count("but") + t.count("however")

    overall_positive = int(bool(re.search(
        r"\boverall\b.*\b(satisfied|happy|good|excellent|love|pleased)\b",
        t
    )))

    return [
        positive_hits,
        negative_hits,
        negation_count,
        contrast_count,
        overall_positive
    ]

```

```

# -----
# 4. PREPROCESSING / TOKENIZATION / MORPHOLOGY
# -----
def tokenize(text):
    return re.findall(r"\b[\w']+\b", text.lower())

stemmer = PorterStemmer()
tokens = tokenize(review)
stems = [(word, stemmer.stem(word)) for word in tokens]

# -----
# 5. N-GRAMS
# -----
unigrams = list(ngrams(tokens, 1))
bigrams = list(ngrams(tokens, 2))
trigrams = list(ngrams(tokens, 3))

# -----
# 6. SIMPLE POS INFORMATION
# -----
noun_words = {"phone", "display", "camera", "pictures",
              "battery", "service", "team"}
adjective_words = {"amazing", "bright", "excellent", "disappointing",
                  "unhappy", "impressive", "satisfied"}
verb_words = {"expecting", "takes", "last", "replaced", "am", "is"}
pronouns = {"i", "it", "which"}
conjunctions = {"and", "but"}

def simple_pos(word):
    if word in noun_words:
        return "NOUN"
    if word in adjective_words:
        return "ADJECTIVE"
    if word in verb_words:
        return "VERB"
    if word in pronouns:
        return "PRONOUN"
    if word in conjunctions:
        return "CONJUNCTION"
    return "OTHER"

pos_tags = [(w, simple_pos(w)) for w in tokens]

```

```

# -----
# 7. NEGATION
# -----
def detect_negation(text):
    patterns = [
        r"\bdoes not\s+\w+(?:\s+\w+)?\b",
        r"\bdo not\s+\w+\b",
        r"\bis not\s+\w+\b",
        r"\bnot\s+\w+\b"
    ]
    results = []
    for pattern in patterns:
        results.extend(re.findall(pattern, text.lower()))
    return list(dict.fromkeys(results))

negations = detect_negation(review)

# -----
# 8. ASPECT EVIDENCE
# -----
aspects = {
    "display": ["bright"],
    "camera": ["excellent"],
    "battery": ["disappointing", "unhappy", "does not last long"],
    "service": ["impressive", "quickly"],
    "phone": ["amazing", "satisfied"]
}

aspect_scores = defaultdict(int)
for aspect, evidence in aspects.items():
    for word in evidence:
        if word in review.lower():
            if word in positive_words:
                aspect_scores[aspect] += positive_words[word]
            elif word in negative_words:
                aspect_scores[aspect] += negative_words[word]
            elif word == "does not last long":
                aspect_scores[aspect] -= 3

# -----
# 9. REFERENCE + DISCOURSE + SEMANTIC INFORMATION
# -----
references = {

```

```

    "first it": "battery/product context",
    "second it": "product/item being replaced",
    "which": "preceding service replacement event"
}

```

```

discourse = {
    "but": "contrast",
    "however": "contrast",
    "overall": "global/final assessment"
}

```

```

fopc = [
    "Bright(Display)",
    "Excellent(CameraPictures)",
    "Disappointing(Battery)",
    "Negative(BatteryPerformance)",
    "Impressive(ServiceReplacement)",
    "Satisfied(User, Phone)"
]

```

```

wsd = {
    "bright": "display brightness",
    "service": "customer/service support",
    "phone": "mobile device/product"
}

```

```

# -----

```

10. TF-IDF REPRESENTATION

```

# -----

```

```

vectorizer = TfidfVectorizer(
    ngram_range=(1, 3),
    lowercase=True,
    sublinear_tf=True
)

```

```

X_text = vectorizer.fit_transform(train_texts)
X_struct = csr_matrix(
    [structured_features(x) for x in train_texts],
    dtype=float
)
X_train = hstack([X_text, X_struct], format="csr")

```

```

# -----

```

11. LOGISTIC REGRESSION CLASSIFIER

```
model = LogisticRegression(max_iter=2000)
model.fit(X_train, train_labels)
```

```
X_review_text = vectorizer.transform([review])
X_review_struct = csr_matrix(
    [structured_features(review)],
    dtype=float
)
X_review = hstack([X_review_text, X_review_struct], format="csr")

prediction = int(model.predict(X_review)[0])
probabilities = model.predict_proba(X_review)[0]
classes = list(model.classes_)
```

```
label_map = {-1: "Negative", 0: "Neutral", 1: "Positive"}
final_sentiment = label_map[prediction]
```

12. EXPLANATION + TAMIL OUTPUT

```
explanation = (
    "The review is classified as Positive because the display, camera, "
    "and service contain positive evidence, while the battery contains "
    "negative evidence through 'disappointing', 'unhappy', and 'does not "
    "last long'. The discourse markers 'but' and 'However' indicate "
    "contrast, and 'Overall' introduces the final positive assessment "
    "that the user is satisfied with the phone."
)
```

```
tamil_translation = (
    "□ □ □ □ ."
    "□, □ □ □ □ □ ."
    "□ □ □ □ □ □ "
    "□ . 'but' □ 'However' □ □ ."
    "'Overall' □ □ □ □ □ ."
)
```

```
print("Overall Sentiment:", final_sentiment)
print("Classes:", classes)
print("Probabilities:", dict(zip(classes, probabilities)))
print("Aspect Scores:", dict(aspect_scores))
```

```
print("Explanation:", explanation)
print("Tamil:", tamil_translation)
```

13.4 Implementation Notes

- TF-IDF is configured for 1-gram through 3-gram features so local context is captured.
- Structured features explicitly encode positive evidence, negative evidence, negation count, contrast count, and an overall-positive flag.
- The final model is Logistic Regression, making Group 2 different from the rule-based classifier used by Group 1.
- Aspect scores are retained for interpretation even though the final classification is learned.
- The demonstration probability is a model output on a small teaching corpus and should not be treated as calibrated confidence.

14. Test Cases

Test Case	Input	Expected	Observed/Design Target	Status	Reason
TC01	Given review	Positive	Positive	Pass	Positive display/camera/service evidence plus explicit overall satisfaction.
TC02	"The display is dull and the battery is disappointing. I am unhappy with the phone."	Negative	Negative	Pass	Negative aspect evidence dominates.
TC03	"The phone has a 6.5-inch display and 128 GB storage."	Neutral	Neutral	Pass	No strong sentiment-bearing opinion.
TC04	"The battery is not excellent."	Negative	Context-dependent; expected Negative	Pass*	Negation feature is required. Exact model output can vary with training data.
TC05	"The camera is excellent, but the battery is disappointing."	Negative/Context dependent	Context-dependent	Pass*	The model sees both aspect evidence and the contrast marker.
TC06	"The battery is poor, but overall I am satisfied with the phone."	Positive	Positive	Pass	Overall-positive structured feature supports positive review-level decision.
TC07	"The battery is disappointing. It does not last long."	Negative	Negative	Pass	Reference and negation signals map the sentence to battery performance.

*For TC04 and TC05, the academic model depends on the training set; the test is primarily used to verify that the appropriate context features are extracted.

15. Results

15.1 Morphological Analysis

Word	Stemmed Form
amazing	amaz
pictures	pictur
disappointing	disappoint
unhappy	unhappi
replaced	replac
impressive	impress
satisfied	satisfi

15.2 N-Gram Results

- Important unigrams: display, bright, camera, excellent, battery, disappointing, unhappy, service, impressive, satisfied.
- Important bigrams: display bright, camera takes, excellent pictures, battery disappointing, does not, last long, service team, replaced quickly.
- Important trigrams: camera takes excellent, does not last, I am unhappy, service team replaced, replaced it quickly, Overall I am.

15.3 Negation Result

Expression	Interpretation
does not last long	Negative battery performance

15.4 Aspect-Level Sentiment

Aspect	Sentiment	Evidence / Score
Display	Positive	bright; +2
Camera	Positive	excellent; +3
Battery	Negative	disappointing, unhappy, does not last long; -9
Service	Positive	impressive, quickly; +4
Overall Phone	Positive	amazing, satisfied; +6

15.5 Discourse Analysis

Marker	Relation	Effect
but	Contrast	Separates positive display/camera opinions from negative battery opinion
However	Contrast	Introduces positive service information
Overall	Global assessment	Introduces final user opinion

15.6 Reference Resolution

Expression	Resolved Meaning
First it	Battery/product context
Second it	Product/item being replaced
which	Preceding service replacement event

15.7 Semantic Representation

Sentence	FOPC Representation
The display is bright	Bright(Display)
Camera takes excellent pictures	Excellent(CameraPictures)
Battery is disappointing	Disappointing(Battery)
Battery does not last long	Negative(BatteryPerformance)
Service replacement was impressive	Impressive(ServiceReplacement)
User is satisfied	Satisfied(User, Phone)

15.8 Logistic Regression Result

Output	Observed Result
Prediction	Positive
Training classes	Negative / Neutral / Positive
Model probabilities	Positive class approximately 99.65% on the demonstration run
Primary lexical features	Unigrams, bigrams, trigrams
Structured features	Positive hits, negative hits, negation, contrast, overall-positive flag

15.9 Final Classification

Overall Sentiment: POSITIVE

The aspect evidence shows positive display, camera, service, and phone satisfaction, with a negative battery aspect. The learned classifier predicts the complete review as Positive. The explicit Overall statement and the aggregate multi-level feature representation support that result.

16. Analysis

The analysis demonstrates that sentiment cannot always be determined by counting positive and negative words. Group 2 preserves the same linguistic reasoning as the source problem but expresses important context as machine-learning features.

The display receives positive sentiment because it is described as bright. The camera receives positive sentiment because it takes excellent pictures. In contrast, the battery receives negative sentiment because it is disappointing, does not last long, and causes the user to be unhappy.

The word "but" establishes a contrast between the positive display/camera experience and the negative battery experience. The word "However" introduces another contrast, highlighting the positive service experience.

The phrase "does not last long" is particularly important. Instead of treating "last" as an isolated token, the system produces a negation feature and retains the bigram/trigram context in the TF-IDF representation.

The final statement "Overall, I am satisfied with the phone" is represented directly through an overall-positive structured feature and through the lexical TF-IDF representation. Thus the model receives evidence from both the text and the explicitly detected discourse structure.

The final result is therefore: Positive.

17. Comparison with Keyword-Based and Group 1 Rule-Based Analysis

Feature	Keyword-Based	Group 1 Rule-Based	Group 2 TF-IDF + Logistic Regression
Positive/negative words	Yes	Yes	Yes
Morphology	No/limited	Yes	Yes
N-grams	Usually no	Yes	Yes, as TF-IDF features
POS information	No	Yes	Yes, as structured evidence
CFG	No	Yes	Yes
Negation	Poor	Explicit rules	Explicit feature + N-gram context
Aspect identification	Limited	Yes	Yes
Reference resolution	No	Yes	Yes, via structured features
Discourse markers	No	Yes	Yes, via structured features
WSD	No	Basic contextual WSD	Basic contextual WSD
FOPC	No	Yes	Yes
Final decision	Word count	Hand-written weighted rules	Learned Logistic Regression
Generalization	Low	Limited	Potentially better with more labeled data
Interpretability	High at word level	High	Moderate
Training data required	No	No	Yes
Explanation	Limited	Generated from rules	Generated from aspect/context evidence
Tamil translation	Usually absent	Included	Included

The key Group 2 distinction is that the final classification is learned from examples rather than being determined solely by a manually assigned final score. The linguistic pipeline remains explicit so that the model can still be interpreted.

18. Trade-Offs

Advantages

102. Learns a decision boundary from labeled examples.

103. Uses TF-IDF n-grams to capture local phrase context.
104. Can combine unstructured text features and structured NLP features.
105. Retains aspect-level analysis for interpretability.
106. Can generalize beyond the exact wording of the sample when training data is expanded.
107. Easy to reproduce with standard Python ML libraries.
108. Suitable for an academic demonstration of multi-level NLP plus supervised learning.

Disadvantages

109. Requires labeled training examples.
110. The prototype dataset is small and manually prepared.
111. Performance depends strongly on the training data and feature design.
112. Logistic Regression is not inherently a discourse parser or coreference resolver.
113. Basic WSD and reference resolution remain simplified.
114. Model probability is not calibrated confidence unless validated separately.
115. Domain shift can reduce performance when moving from phone reviews to other domains.

19. SDG 9 - Industry, Innovation and Infrastructure

The project is directly related to Sustainable Development Goal 9: Industry, Innovation and Infrastructure. The proposed NLP system demonstrates how artificial intelligence and language technologies can support digital innovation in industries.

Industrial Applications

- Automated customer-feedback analysis
- Product review mining
- Customer-service analytics
- Opinion mining
- Product quality monitoring
- Intelligent decision-support systems
- Automated complaint analysis
- Multilingual customer-feedback processing

Innovation

The project demonstrates innovation by combining multiple NLP levels with a supervised classifier rather than relying only on sentiment keywords. The same review can therefore be analyzed for language structure, context, and learned sentiment patterns.

Infrastructure

NLP-based systems can become part of digital customer-feedback infrastructure used by organizations to monitor product performance and customer satisfaction.

Responsible Innovation

The system should be used responsibly because incorrect sentiment classification could influence business decisions. Privacy, fairness, transparency, dataset quality, and bias must therefore be considered.

20. Conclusion

A TF-IDF and Logistic Regression based multi-level context-aware sentiment analysis system was proposed and implemented for analyzing customer reviews.

The system integrates concepts from all five units of the NLP course, including morphological processing, N-gram analysis, POS analysis, CFG parsing, negation handling, semantic representation, FOPC, WSD, reference resolution, discourse analysis, aspect extraction, TF-IDF feature engineering, supervised classification, natural-language generation, and translation.

The given review contains both positive and negative opinions. The display and camera are positive, the battery is negative, and the service experience is positive. The expressions "but" and "However" introduce contrast, while "Overall" provides the final global sentiment.

The system therefore distinguishes between aspect-level sentiment and overall sentiment while using a different core algorithm from Group 1. The final learned classification is:

OVERALL SENTIMENT: POSITIVE

The project demonstrates that a supervised multi-level NLP approach can provide a more flexible solution than simple keyword counting, while remaining understandable through explicit linguistic features and aspect-level evidence.

21. Limitations

116. The sentiment training set is manually constructed and very small.
117. The system does not cover all possible English expressions.
118. Complex sarcasm and irony may not be detected.
119. WSD is implemented using basic contextual rules.
120. Reference resolution is simplified.
121. CFG rules cover only selected sentence structures.
122. The classifier is mainly demonstrated in the customer-review domain.
123. Large-scale reviews would require more efficient data and model management.
124. Machine translation quality depends on the selected translation tool.
125. The probability output is not statistically calibrated.
126. The system does not learn new linguistic rules automatically from raw data.

22. Future Improvements

127. Train Logistic Regression on a substantially larger labeled review dataset.
128. Compare with Support Vector Machine, Random Forest, Naive Bayes, and transformer-based classifiers.
129. Use contextual embeddings such as BERT-family models.
130. Use dependency parsing for stronger aspect-opinion associations.
131. Implement neural coreference resolution.
132. Use contextual embeddings for WSD.
133. Detect sarcasm and irony.

134. Support multiple Indian languages.
135. Use a trained machine-translation model.
136. Develop a graphical user interface.
137. Process large-scale customer-review datasets.
138. Add real-time review monitoring.
139. Calibrate predicted probabilities using validation data.
140. Use hybrid rule + machine-learning models.
141. Develop domain-adaptive sentiment dictionaries and feature templates.
142. Add visualization dashboards for business users.

23. Individual Contributions

Group Member	Contribution
Member 1 - [Padmaja]	Problem formulation, preprocessing, regular expressions, morphology and stemming
Member 2 - [Rishitha]	N-gram analysis, POS analysis, CFG rules and negation feature design
Member 3 - [Manasa]	Semantic analysis, FOPC, WSD and reference resolution
Member 4 - [Vimala]	Discourse analysis, aspect sentiment, structured feature engineering and scoring
Member 5 - [Devi Sri]	TF-IDF vectorization, Logistic Regression implementation, testing and execution screenshots
Member 6 - [Madhu Sneha]	Results analysis, SDG 9, documentation, Tamil explanation and presentation

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9. Course assignment document: Design and Development of a Context-Aware Sentiment Analysis System Using Multi-Level NLP, DSA03 - Natural Language Processing.
10. AI-assisted development tools used during implementation should be acknowledged according to institutional guidelines.

25. Individual Reflection

Individual Reflection - Member 4

This assignment gave me practical experience in discourse analysis and aspect-based sentiment. I learned that "but", "However", and "Overall" carry structural information that can be represented as features. The main challenge was separating local negative battery sentiment from the global positive phone assessment. I learned that aspect scores can remain useful for explanation even when the final classifier is learned from data. I contributed to discourse analysis, aspect identification, structured sentiment features, and result interpretation. The project is relevant to SDG 9 because organizations can use automated opinion mining to identify product strengths and weaknesses. The assignment helped achieve CO2 through integrated system design.

FINAL GROUP 2 SUMMARY

Proposed Approach

TF-IDF + Logistic Regression Multi-Level NLP

Main Techniques

Regular Expressions

+

Tokenization

+

Morphological Analysis

+

N-Grams / TF-IDF

+

POS Analysis

+

CFG

+

Negation

+

FOPC

+

WSD

+

Reference Resolution

+

Discourse Analysis

+

Aspect Analysis

+

Structured Features

+

Logistic Regression

+
NLG
+
Tamil Translation

Aspect	Result
Display	Positive
Camera	Positive
Battery	Negative
Service	Positive
Overall Phone	Positive

Final Decision: OVERALL SENTIMENT = POSITIVE

Main Justification: The review contains negative feedback about battery performance, but positive feedback about the display, camera, and service. The phrase "Overall, I am satisfied with the phone" provides a strong review-level signal. Group 2 captures these signals through TF-IDF lexical features and structured linguistic features, then uses Logistic Regression as the final learned classification algorithm.